

The UAT Physical Guarantee of Navier-Stokes Regularity

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Framework: Unified Applied Time (UAT) — Loop Quantum Gravity base

Original Date: 26 Oct 2025 — Updated: 9 Sep 2026 (Addenda v9.1 vs OpenAI)

Related DOIs: 10.5281/zenodo.17729221 (UAT) | 10.5281/zenodo.18210808 (UPC) | 10.5281/zenodo.22678424 (Antifrequency)

GitHub: <https://github.com/miguelpercu/The-UAT-Physical-Guarantee-of-Navier-Stokes-Regularity>

***IMPORTANT CLARIFICATION:** This repository does NOT claim a formal mathematical proof in the sense of the Clay Millennium Prize (pure incompressible Navier-Stokes PDE without cutoffs). It provides the **physical guarantee** that regularity must hold in nature due to quantum-gravitational constraints. This distinction became crucial after OpenAI's announcement on Sep 8, 2026.*

1. Abstract

This document presents the physical foundation for resolving the Navier-Stokes existence and smoothness problem through the Unified Applied Time (UAT) framework. We demonstrate that quantum gravitational constraints, specifically the Planck-scale energy density limit, inherently bound fluid velocity singularities. While not a formal mathematical proof, this work establishes the physical guarantee that such a proof must exist for the **physically regularized** equations, as infinite velocity singularities violate fundamental quantum spacetime structure.

Output (corrected):

- $\rho_{\text{max}} = M_{\text{Planck}} / L_{\text{Planck}}^3 \approx 5.155e96 \text{ kg/m}^3$
- $|v|_{\text{max}} = \sqrt{2 \rho_{\text{max}} c^2 / \rho_{\text{fluid}}} \approx 9.6e56 \text{ m/s}$ (with c^2) or $1.015e47 \text{ m/s}$ (code v1 without c^2 , both finite)

Finitude is the key insight: mathematical singularities require infinite energy density, which violates UAT/Planck bound.

2. Background: The Millennium Problem

The Clay problem asks:

$$\begin{aligned} 1 \quad & \partial \mathbf{v} / \partial t + (\mathbf{v} \cdot \nabla) \mathbf{v} = -1/\rho \nabla p + \nu \nabla^2 \mathbf{v} + \mathbf{f} \\ 1 \quad & \nabla \cdot \mathbf{v} = 0 \end{aligned}$$

Starting from smooth initial conditions, does solution remain smooth (C^∞) for all $t > 0$, or can it develop finite-time blow-up (Cases C/D)?

For 90 years, Leray (1934) weak solutions existed, but classical regularity in 3D remained open. The suspicion was that viscosity $\nu \nabla^2 \mathbf{v}$ would prevent blow-up.

3. What OpenAI Announced on Sep 8, 2026

On Sep 8, 2026, OpenAI announced a solution claiming blow-up (finite-time singularity) using:

- 1000 AI agents on Euler (50h) then 10,000 agents on Navier-Stokes (11h), total ~88h, 130B tokens, \$15M estimated cost
- Formal verification in Lean (17h)
- Mechanism: vortical filament spiraling inward with axial stretching, inertia, pressure and viscosity cancel leaving net smooth forcing while pointwise velocity diverges
- Total energy remains finite because concentration volume $\rightarrow 0$, but local energy density $\rightarrow \infty$
- Controversy: Tristan Buckmaster / Levent Alpöge had announced stepping stones hours before; OpenAI denies using their Codex data

- Clay President Martin Bridson: evaluation is deliberately unhurried and absolutely rigorous. No prize awarded yet.

Key point: OpenAI's result, if verified, shows **pure continuum mathematics allows blow-up**.

4. What UAT Provides: The Physical Cutoff

UAT, rooted in Loop Quantum Gravity, imposes:

Theorem (UAT Energy Density Bound):

$$\begin{aligned} 1 \quad \rho_{\max} &= M_{\text{Planck}} / L_{\text{Planck}}^3 \approx 5.155e96 \text{ kg/m}^3 \\ 1 \quad v_{\min} &= L_{\text{Planck}}^3 \end{aligned}$$

Kinetic energy density:

$$\begin{aligned} 1 \quad \epsilon_{\text{kin}} &= 1/2 \rho_{\text{fluid}} |v|^2 \leq \rho_{\max} c^2 \\ 1 \quad \Rightarrow |v|_{\max} &= \sqrt{2 \rho_{\max} c^2 / \rho_{\text{fluid}}} \end{aligned}$$

For water $\rho_{\text{fluid}}=1000 \text{ kg/m}^3$:

- $v_{\max}$ (with c^2 , physically correct): $\sim 9.6e56 \text{ m/s}$
- $v_{\max}$ (v1 code without c^2): $1.015e47 \text{ m/s}$

Both finite. Therefore:

Theorem (UAT Smoothness Guarantee): Quantum spacetime discreteness guarantees solutions to the physically regularized Navier-Stokes equations remain smooth, as singularities would require infinite energy density violating Planck bound.

Proof by contradiction: Assume singularity at $t_c \Rightarrow |v| \rightarrow \infty \Rightarrow \epsilon_{kin} \rightarrow \infty$ violates $\epsilon_{kin} \leq \rho_{max} c^2$. Contradiction.

5. Relation: Why Both Can Be True

Aspect	OpenAI (Sep 2026)	UAT Guarantee (This Repo)
Domain	Pure math, continuum, no cutoff	Physical reality, LQG cutoff V_{min}
Equation	Original Navier-Stokes	Navier-Stokes + Planck bound / Ψ -field
Conclusion	Blow-up possible mathematically	Blow-up impossible physically
Energy density	Local $\rightarrow \infty$ allowed in math	Local $\leq \rho_{max} c^2$ forbidden in physics
Implication	PDE is effective theory, carries seed of its own limit	Nature regularizes via quantum geometry

This is exactly the fly/human filter from `Note_03_08_2026.pdf` : Trinity College Dublin 2013 *Musca domestica* 200–250 Hz vs human 50-60 Hz. Human 50 Hz sees blurred superposition >70% ambiguity, fly 200 Hz sees discrete deterministic steps <20%. Both correct in frame. Question that emerges: What is real time when no one measures it? Answer UAT: Real time anchored not to human second but to causal membrane Bit0/Bit1 geometry $R_{geom}=0.279182$, $k_{early}=0.967$, $\kappa_{crit}=10^{-78}$.

Navier-Stokes blow-up is same class: continuum without observer filter blows up; with Planck filter, finite.

6. Python Implementation

python

```
1 from scipy.constants import c, G, hbar
1 import numpy as np
1
1 class UAT_NavierStokes_Restriction:
1     def __init__(self):
1         self.c, self.G, self.hbar = c, G, hbar
1         self.gamma = 0.2375 # Barbero-Immirzi
1         self.rho_fluid = 1000.0
1
1     @property
1     def L_PLANCK(self): return np.sqrt(self.hbar*self.G/self.c**3)
1     @property
1     def M_PLANCK(self): return np.sqrt(self.hbar*self.c/self.G)
1     def RHO_MAX_UAT(self): return self.M_PLANCK / self.L_PLANCK**3
1
1     def calculate(self):
1         rho_max = self.RHO_MAX_UAT()
1         v_max = np.sqrt(2*rho_max*self.c**2/self.rho_fluid) # correct
1         print(f"p_max: {rho_max:.3e} kg/m³")
1         print(f"|v|_max: {v_max:.3e} m/s finite => smooth")
1         return v_max
```

See `limitación_del_caos_de_Navier_Stokes.py` for full version.

7. Files in This Repo

- `Navier_Stokes.pdf` — LaTeX paper (Oct 26 2025, 5 pages)
- `limitación_del_caos_de_Navier_Stokes.py` — UAT restriction calculation
- `Note_03_08_2026.pdf` — Before the Metrics: Observer's Dilemma (fly/human filter origin)
- `Addenda_v9_1_NavierStokes_Rgeom.pdf` — Addenda Sep 9 2026 linking Navier-Stokes blow-up to R_geom regularization
- `supplementary_script.py` — Simulation human 50Hz vs fly 200Hz observing 1000Hz quantum oscillation

8. Roadmap to Formal Proof

UAT provides **why**, mathematicians must provide **how**. Remaining tasks to formalize physical constraint within PDE:

- Modified viscosity terms at small scales: $v_{\text{eff}}(l) = v * (1 + (L_P/l)^\alpha)$
- Regularization via Schrödinger-Poisson phase field (see Zenodo 2026-02-27 regularization)
- Energy dissipation via coherence field Ψ : $-\gamma_c \Psi(t) u$ term (Ψ -NSE v1.0)

9. Citation

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1 Percudani, M.A. (2025). The UAT Physical Guarantee of Navier-Stokes R
1 GitHub: https://github.com/miguelpercu/The-UAT-Physical-Guarantee-of-
1 Zenodo (when restored): To be linked to 10.5281/zenodo.17729221 evolu
1 ORCID: 0009-0007-1748-3212
```

10. References

- Clay Mathematics Institute. Navier-Stokes Existence and Smoothness Millennium Problem.
 - Rovelli, C. Quantum Gravity (2004).
 - OpenAI (Sep 8 2026). An OpenAI model proposes a solution to the Navier-Stokes problem.
 - New Scientist (Sep 9 2026). OpenAI has solved the Navier-Stokes Millennium problem using \$15m of AI effort.
 - Buckmaster, T. et al. (Sep 2026) Euler blow-up stepping stones.
 - Percudani, M.A. (2026). Note_03_08_2026 Before the Metrics; Technical_Note Unified Action.
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This is physical guarantee, not Clay formal proof. If OpenAI blow-up is confirmed, this repo becomes the physical regularization that nature uses to avoid mathematical singularity — same role as $R_{\text{geom}}=0.2791$ truncation at 348.12° leaving 3.3% arrow-of-time deficit.